A Proposal for a Master's Thesis

To be Entitled

______~~HIV~~ PrEP Data Analysis and Reporting Tool ~~Indicator Reporting~~ (PrEP-DART~~HIR~~)~~:~~
_____A d~~D~~ata Pipeline developed on Health Data Lab Frameworks
by

[Gokul Thothathri]

To Be Submitted in Fulfillment for the Degree of MASTER OF SCIENCE BIOINFORMATICS

21~~01~~ - 01~~16~~ - 2025~~4~~

Approved: ________________________ Date: ______________
(Thesis Advisor/Examiner 1)

Approved: ________________________ Date: ______________
(Supervisor/Examiner 2)

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1 Title

PrEP Data Analysis and Reporting Tool (DART): A Data Pipeline Developed on Health Data Lab Frameworks~~**HIV-PrEP Indicator Reporting (HIR): A Data Pipeline developed on Health Data Lab Frameworks**~~

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2 Abstract

HIV pre-exposure prophylaxis (PrEP) programs are essential in preventing the transmission of HIV, requiring efficient monitoring and reporting frameworks to assess their reach and impact. ~~HIV pre-exposure prophylaxis (PrEP~~This master thesis~~)~~ aims to develop a robust and scalable data analysis pipeline for reporting HIV-PrEP indicators, leveraging anonymized health data and aligning with the indicator framework defined by the European Centre for Disease Prevention and Control (ECDC) [1]. Using anonymized datasets ~~generated~~ provided by the Health Data Lab, the methodology automates data cleaning, integration, and transformation processes to compute and report key metrics. ~~The proposed pipeline follows a structured workflow, incorporating advanced anonymization techniques, including k-anonymity and pseudonym replacement, to ensure privacy compliance.~~ It integrates relational data across multiple tables, as defined by the dataset's entity-relationship model (ER Diagrams), and maps indicators to relevant fields, such as patient demographics, prEP prescriptions, adherence records, and laboratory results, ~~-~~ for computational calculation [2]. The methodology outputs comprehensive reports, offering insights into program performance while remaining adaptable to future extensions or real-world data. This master thesis ~~e~~ project ~~proposal~~ aims to address the critical challenges in public health informatics, such as ~~including the ethical handling of sensitive data and~~ the automation of complex reporting tasks, by focusing on the development of scalable data analysis pipeline. By reducing the time and effort required for manual reporting, the proposed pipeline not only supports efficient monitoring of HIV-PrEP programs but also serves as a blueprint for scalable and privacy-preserving data solutions in healthcare [3]. The results are expected to contribute to public health decision-making and improve resource allocation for HIV prevention strategies.

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3 Introduction

Health Data: Resources and Tools from FDZ Gesundheit (DaTrav Structure)

The Health Data Lab (FDZ Gesundheit) is a research infrastructure that provides access to anonymized health-related datasets collected from healthcare providers, insurance claims, and public health institutions in Germany. These datasets are curated to support research in public health, epidemiology and health informatics while ensuring strict compliance with data protection regulations. The Health Data Lab (FDZ Gesundheit) provides descriptions of resources and documentation to support researchers in accessing and analyzing health-related datasets. FDZ Gesundheit offers a detailed description of the available datasets, outlining their structure, variables, and interrelationships. A dedicated platform includes comprehensive information about the datasets, such as variable definitions, coding systems, and data formats. To assist

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Commented [SD8]: I think first of all you need to describe shortly what FDZ data is and were it comes from. Imagine someone never heart of FDZ. What is the basis of the data, what are the ways FDZ gets data? Then go deeper into what's on the website.

researchers in understanding the data architecture, the site provides Entity-Relationship (ER) diagrams, which visually map the relationships between tables, facilitating efficient data integration and analysis. Additionally, the main FDZ Gesundheit website explains the broader context and purpose of the data lab, emphasizing its role in offering secure access to health data for research. The site highlights the measures taken to ensure data anonymization, including techniques such as k-anonymity, pseudonym replacement, and selective sampling, which preserve privacy while maintaining data utility. This ensures that sensitive health data can be analyzed ethically and responsibly. Furthermore, FDZ Gesundheit focuses on enabling data-driven public health research by providing secure and well-documented data pipelines for analysis. These combined resources empower researchers to work confidently with complex health datasets, allowing for robust analyses and informed decision-making. Together, the documentation and tools provided by FDZ Gesundheit serve as a foundational database for advancing public health research in compliance with privacy and ethical standards.

PrEP Indicators: Measuring the Impact of HIV Prevention Strategies

The European Centre for Disease Prevention and Control (ECDC) has established a comprehensive framework for monitoring HIV pre-exposure prophylaxis (PrEP) programs across the EU/EEA [6]. This framework includes indicators that are categorized into three domains: pre-uptake, uptake and coverage, and continued and effective use. Pre-uptake indicators evaluate factors influencing PrEP adoption, such as service availability, awareness among potential users, and willingness to use ~~PrEP.~~prEP Uptake and coverage indicators measure the actual utilization and reach of PrEP, focusing on metrics like the number of current and new PrEP users, overall PrEP coverage, and the PrEP-to-Need Ratio (PnR), which serves as a proxy for assessing coverage. Continued and effective use indicators examine the sustainability and effectiveness of PrEP over time, tracking metrics like recent PrEP use among individuals newly diagnosed with HIV and the continuation rates of users after 12 months. These indicators are further classified into core, supplementary, and optional categories, based on their importance and feasibility for regular monitoring. Core indicators are essential and feasible for routine use, supplementary indicators provide valuable insights but may require additional resources, and optional indicators offer deeper understanding but are less practical for standard reporting. By implementing this structured approach, PrEP program performance can be systematically assessed, gaps identified, and strategies optimized to enhance HIV prevention efforts across diverse populations. This robust monitoring framework enables data-driven decision-making to advance public health objectives. Monitoring will be vital to understand whether programs succeed in engaging people into using PrEP appropriately, and in reducing the HIV epidemic. [5]

PrEP Surv: A Comprehensive Initiative for Monitoring HIV Prevention in Germany

The project “Surveillance of HIV pre-exposure prophylaxis provision within the SHI system in Germany (PrEP Surv ~~(PrEP Surveillance~~), initiated by Robert Koch-Institute and financed by the German Federal Ministry of Health (BMG), represents a pivotal effort in the surveillance and evaluation of HIV pre-exposure prophylaxis (PrEP) among HIV-negative populations in Germany. Building on the outcomes of the prior project, "Evaluation of the Introduction of HIV Pre-Exposure Prophylaxis as a Benefit of Statutory Health Insurance~~,~~ (EvE-PrEP)", PrEP Surv wa~~i~~s designed to monitor the efficacy and reach of PrEP within the framework of statutory health insurance (~~STI~~SHI). The initiative encompasses key components of PrEP care, including the administration of [tenofovir disoproxil + emtricitabine, TDF/FTC] [4], patient counseling, medication supply, and associated medical precautions such as diagnostic follow-ups. The launch of PrEP Surv ~~marks~~ marked the translation of evaluation findings into a structured surveillance program, aligned with EU-PrEP monitoring guidelines. A primary focus of the initiative wa~~i~~s to assess the incidence of HIV and sexually transmitted infections(STIs) among the prEP users in Germany ~~development of HIV and sexually transmitted infections (STIs) and HIV among PrEP users in Germany~~. Project ~~This effort necessitates the integration of extensive datasets sourced from diverse entities, including pharmacies, health insurance providers, HIV specialty care centers and the Health Data Lab (FDZ).~~ p~~P~~rEP Surv operated~~s~~ across three interconnected modules: (1) secondary data collection from pharmacies, health insurers, and FDZ, (2) enhancement of data collection from German HIV specialty care centers within the dägna network to strengthen PrEP service reporting, and (3) collaboration with commun~~al bod~~ities with a potential need for PrEP to identify barriers to PrEP access and address unmet needs in HIV prevention. By leveraging these comprehensive data streams, PrEP Surv aimed~~s~~ to advance the penetration of PrEP services and establish an evidence-based foundation for future HIV monitoring strategies in Germany.

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4 Objectives

1. To develop a robust data pipeline, named PrEP-DART ~~HIR~~ (prEP- Data Analysis and Reporting Tool~~HIV-PrEP Indicator Reporting~~), for automating data processing and reporting.
~~2. To integrate datasets from the Health Data Lab (FDZ) into a unified framework using entity-relationship (ER) diagrams.~~
~~3. To implement advanced anonymization techniques, such as k-anonymity and pseudonym replacement, ensuring compliance with privacy standards.~~
~~4. To automate data cleaning, integration, and transformation processes to prepare datasets for analysis.~~
~~5.~~2. To map ECDC-defined PrEP indicators to the Health Data Lab datasets for computational calculation. (Core, supplementary and optional PrEP indicators, such as PrEP coverage, new PrEP Users, and PrEP-to-Need Ratio (PnR), in alignment with EU guidelines and ensuring the reporting system is adaptable for real-world data and future extensions.
~~6.~~3. Validation of HIR pipeline's performance using mock datasets from FDZ, ensuring accuracy and efficiency in indicator computation.
~~7. Provide scalable blueprint for similar data-driven public health initiatives.~~

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Commented [MT14]: I agree, at least not the core focus.

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5 Literature Review

1. A Review of Anonymization for Healthcare Data [7]
2. Monitoring HIV pre-exposure prophylaxis programs in the EU/EEA (ECDC, 2024)
3. Algorithms to anonymize structured medical and healthcare data: A systematic review [8]
4. A richly interactive exploratory data analysis and visualization tool using electronic medical records [9]
5. Effective Environmental Public Health Surveillance Programs: A Framework for identifying and evaluating data resources and indicators [10]
6. Interactive Visualization Applications in Population Health and Health Services Research: Systematic Scoping Review [11]

6 Methodology

6.1 Data Collection

The master thesis project utilizes synthetically generated datasets to simulate real-world data while preserving privacy and confidentiality requirements. The synthetic data is created using the framework provided by the Health Data Lab (FDZ Gesundheit). The Public Use File (PUF) repository enables researchers to develop and test analysis pipelines with datasets that mimic the structure and schema of FDZ's actual datasets, including entity-relationship (ER) models and variable specifications. The PUF data is modeled to replicate the relationships between various entities in the FDZ datasets. These include patient demographics, prescription data, healthcare service utilization and regional health statistics. The data structure is based on ER diagrams that define the relationship between the tables, making it suitable for developing and testing relational databases. The synthetic nature of PUF data ensures privacy by destroying correlations, effectively preventing the extraction of sensitive insights or interferences that could compromise confidentiality. While the dataset will be structured like real-world data, it does not maintain the underlying relationship or patterns that exists in the actual data. This complementary nature of PUF data helps researchers focus on building robust analytical solutions without compromising privacy or requiring access to sensitive information.

~~PrEP Surv will gather data from diverse sources to ensure a comprehensive understanding of PrEP usage and its impact. Secondary data will be collected from pharmacies, the statutory health insurance (SHI) system, and the FDZ Health Data Lab, providing a robust quantitative foundation. This will be complemented by the integration of service-specific data from HIV specialty care centers within the dägna network, offering insights into the operational aspects of PrEP provision. To address the social and systemic barriers to PrEP adoption, qualitative data will be collected from communal bodies. This multi-faceted approach will enable a holistic evaluation of PrEP adoption, effectiveness, and accessibility within Germany's healthcare system. PrEP indicators, as defined by the European Centre for Disease Prevention and Control (ECDC), are essential metrics for monitoring the reach, adoption, and effectiveness of HIV pre-exposure prophylaxis (PrEP) programs. These indicators are categorized into three domains: pre-uptake, uptake and coverage, and continued and effective use. To calculate these indicators, comprehensive data collection is required, particularly focusing on quantitative and qualitative datasets that align with the indicator framework. The Health Data Lab (FDZ Gesundheit) plays a critical role in providing the necessary datasets for calculating these indicators. The FDZ offers anonymized datasets that include detailed records on patient demographics, prescription data, healthcare service utilization, and regional health statistics. These datasets are structured using entity-relationship (ER) diagrams, which map the relationships between various tables, ensuring seamless integration and analysis. Data collected from the Health Data Lab supports the computation of core PrEP indicators such as: PrEP Coverage: The proportion of individuals using PrEP relative to the population at risk. PrEP Uptake: The number of current and new PrEP users during a given period. PrEP-to-Need Ratio (PnR): A comparison of PrEP usage to the number of new HIV diagnoses, serving as a proxy for program reach. Continuation Rates: The proportion of users who remain on PrEP after a year of initiation. The collected data is subjected to advanced anonymization techniques, including k-anonymity and pseudonym replacement, to safeguard patient privacy. These privacy-preserving measures enable the Health Data Lab to provide datasets that are both secure and analytically robust. By leveraging the structured datasets from FDZ and aligning them with ECDC-defined PrEP indicators, this~~

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~~project ensures a data-driven approach to understanding and enhancing PrEP adoption, effectiveness, and accessibility across Germany.~~

6.2. Pipeline Architecture: prEP-DART~~HIR~~ (prEP- Data Analysis and Reporting Tool~~HIV-PrEP Indicator Reporting~~)

The development of the *prEP-DART (prEP- Data Analysis and Reporting Tool)* ~~HIV-PrEP Indicator Reporting (HIR)~~ pipeline involves creating a robust, automated system for processing and analyzing health datasets to compute key PrEP indicators. This process includes multiple stages, each designed to ensure the pipeline is scalable, efficient, and compliant with privacy standards. The pipeline begins with dataset models~~s~~ provided by the Health Data Lab (FDZ Gesundheit), which serves as the foundation for the PUF generated data~~all subsequent processes~~. ~~These datasets are well-structured and anonymized, containing patient demographics, prescription records, and healthcare service utilization data. Entity-relationship (ER) diagrams are used to map the interconnections between tables, ensuring accurate integration and analysis. The second step involves aligning the datasets with PrEP indicators defined by the European Centre for Disease Prevention and Control (ECDC).~~ Further, ~~This step~~ mapping of ~~s~~ relevant fields in the generated data ~~FDZ datasets~~ to the ECDC-defined indicators for accurate calculations~~. The anonymized datasets are further processed into Public Use Files (PUFs). These PUFs are generated using privacy-preserving techniques such as k-anonymity and pseudonym replacement to ensure compliance with data protection standards.~~ The Public Use File (PUFs) serve as an intermediary dataset, allowing the pipeline to operate on secure and anonymized information without compromising analytical utility. At the core of the architecture is the *prEP- Data Analysis and Reporting Tool*~~HIR~~ pipeline, which automates: (1) Data cleaning and transformation ~~(2) Integration~~ of Public Use File (PUF) datasets ~~into a unified framework~~ (2~~3~~) Computational calculations for key indicators like prEP coverage, uptake, and the prEP-to-Need Ration (PnR) (3) Validation and testing with different mock PUF datasets. ~~The pipeline processes the PUF-generated data to compute the required PrEP indicators. This ensures privacy compliance while enabling accurate calculation and analysis of metrics. Mock datasets are used during development to validate the pipeline's performance before applying it to live data .~~ The pipeline outputs tabular summaries and visualizations, created using Python libraries such as Matplotlib, Seaborn, and Plotly. These outputs are not designed for real-world public health monitoring but rather serve as benchmarks for demonstrating the pipeline's functionality and readiness for future integration with real-world datasets.~~The pipeline outputs comprehensive reports in the form of (1) Tabular outputs (2) Visualizations, these reports are essential for public health decision-making, providing insights into the performance of PrEP programs, gaps in implementation, and areas requiring improvement .~~ This architecture demonstrates how the *prEP- Data Analysis and Reporting Tool* pipeline ~~HIR pipeline~~ integrates with PUF~~diverse~~ datasets, aligns them with standardized PrEP indicators, and transforms anonymized data into actionable insights. By leveraging privacy-preserving methods and automated workflows, the pipeline lays the foundation for ~~ensures~~ efficient and ethical reporting and validating~~e~~ to support public health monitoring and improve resource allocation for HIV prevention strategies.

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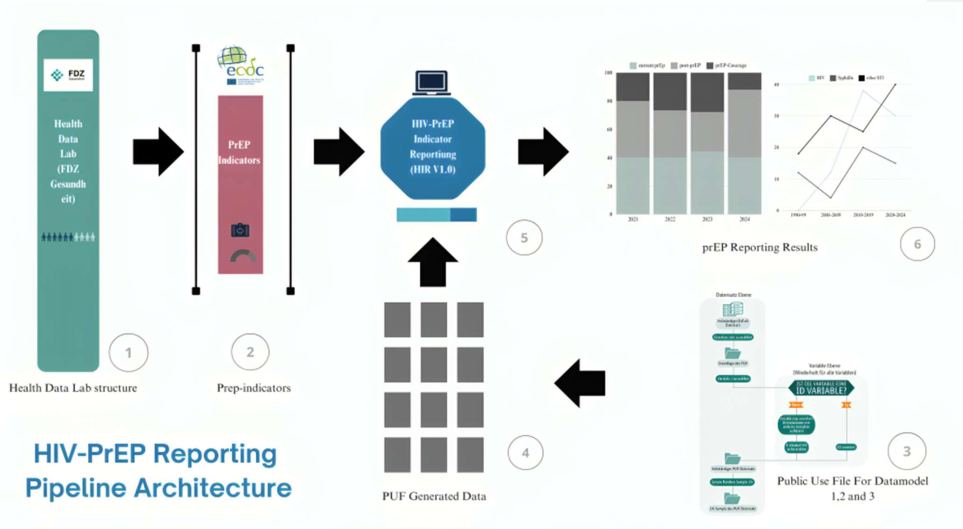

6.3 *Data Analysis*

The project will assess its impact through key metrics aligned with the ECDC-defined indicators, such as PrEP adoption rates, coverage, and the PrEP-to-Need Ratio (PnR).~~The project will assess its impact through key metrics, including the adoption rates of PrEP and the incidence of HIV and sexually transmitted infections (STIs) among PrEP users. These metrics will provide crucial insights into the effectiveness and reach of PrEP services within the healthcare system. The HIR (HIV-PrEP Indicator Reporting) Algorithm will be designed to automate the processing, integration, and analysis of health data to compute key PrEP indicators. The pipeline takes in anonymized dataset structure from the Health Data Lab (FDZ Gesundheit), which include patient demographics, prescription records, and healthcare service data. The data is generated using Public Use File protocol, where each variable in the dataset is independently sampled without replacement to break any relationships or correlations with other variables ensuring that all the univariate distributions and errors are preserved whereas all variable correlations are eliminated. The protocol uses anonymization techniques such as k-anonymity and pseudonym replacement where the identifiers (e.g., patient IDs) are replaced with randomly generated pseudonyms using pools of valid identifiers.~~ Overall the PUF structure maintains the same structure as the DaTrav structure with core components of Entity-Relationship (ER) models, tables and variables, Data models and structural components such as Primary Keys (PK), relationships, Hierarchical Variables, Anonymized data and standardization. The data pre-processing steps include handling missing or inconsistent values, standardizing formats across datasets (date formats, variable names) followed by removing duplicates and irrelevant records. Secondly, the algorithm uses ER diagrams to integrate relational datasets, ensuring consistency in linking tables such as patient records, prescriptions, and service data. Further the algorithm maps relevant data fields to the core, supplementary and optional indicators defined by the European Centre for Disease Prevention and Control (ECDC). ~~Key indicators include: prEP coverage, prEP uptake, prEP-to-Need Ration (PnR) and continuation~~

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~~rates.~~ These computations are performed for specific timeframes (eg., annually and quarterly) to monitor trends. The validations will be performed using running test cases on mock datasets to ensure accuracy and comparing calculated metrics against benchmark values and historical data. The data analysis will be distinguished as (1) Descriptive Analysis, to provide an overview of the dataset and key metrics (2) prEP-Trend analysis, to monitor changes in prEP indicators over time (3) Comparative Analysis, to identify disparities in prEP adoption across demographics or regions using group-wise comparisons (e.g., by age, gender, geography) (4) Predictive Insights, to forecast future trends in prEP adoption and its impact on historical data (5) Indicator-Specific Insights, to identify gaps in service provisions(prEP-coverage), to highlight the effectiveness of the program in reducing new infections(PnR) and to evaluate adherence and retention among prEP users(continuation rates). The resulting insights empower beneficiaries to make data-driven decisions, optimize resource allocation, and enhance the overall impact of HIV prevention programs. This pipeline ~~algorithm~~ serves as a validation blueprint for scalable and privacy-compliant data solutions in public health informatics in the future.

7 Timeline

Phase	Duration	Deliverables
Literature Review	3 weeks	Annotated bibliography, draft background on PrEP indicators, privacy techniques
Data Collection	2 weeks	Validation of FDZ Dataset Integration and PUF Generation~~Integration of FDZ datasets, PUF generation validation~~
Framework Development	2 months	A functional prototype of the data pipeline capable of processing synthetic PUF datasets and computing key PrEP indicators.~~Interoperable data pipeline prototype with initial anonymization and integration~~
Indicator Mapping	3 weeks	Mapping of ECDC-defined indicators to the PUF~~FDZ~~ datasets
Data Analysis	1 month	Statistical models, calculations of PrEP indicators, and initial visualizations
Automated Reporting	2 weeks	Reports with tables and visualizations, final testing
Validation and Testing	2 weeks	Validation with different mock datasets, accuracy testing

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Thesis Writing	1 month	Final thesis including methodology, results, and discussions

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